happening to Schrödinger's cat before you look in the box, insisting only that it's meaningless to talk about the unobservable. But, in Bohm's pilot-wave interpretation, not only can you ask but there's an answer: before you look in the box, the cat is either dead or alive, and opening the box merely reveals which is true. The act of observation has nothing to do with the condition of the cat.

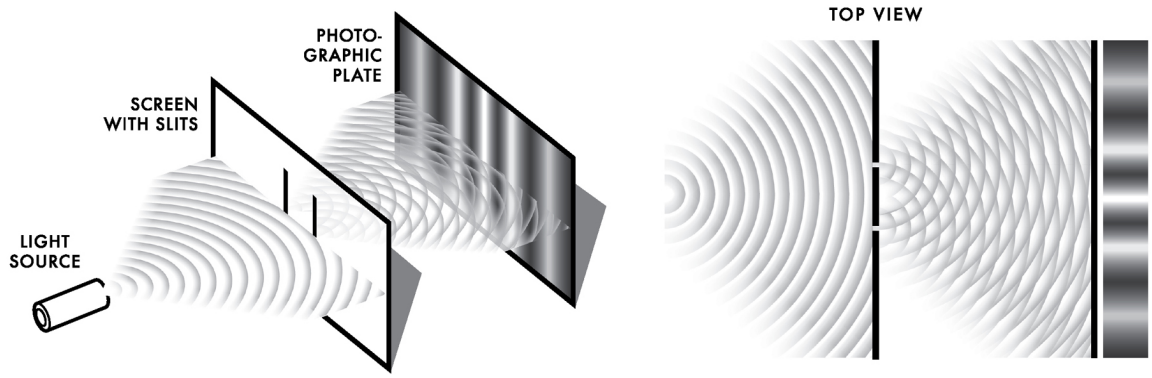


Figure 5.2. Waves in the double-slit experiment interfere with each other.

At first blush, this seems far too easy. If, in Bohm's theory, there is nothing strange about particle positions or Schrödinger's cat, how can it possibly hope to reproduce all the bizarre results of quantum physics? But the guarantee is in the mathematics: Bohm's theory is mathematically equivalent to the Schrödinger equation, the central equation of quantum physics, and therefore it must make the same predictions as any other interpretation. This is technically true, but it doesn't give a feel for how Bohm's interpretation actually works. To do that, we'll need to look at one of the strangest experiments in all of quantum physics: the double slit.

The great physicist Richard Feynman famously said that the double-slit experiment "has in it the heart of quantum mechanics," and that "in reality, it contains the *only* mystery." Yet, for all that hype, it's a surprisingly simple experiment. Set up a screen in front of a photographic plate, and place two narrow closely spaced slits in the screen. Then shine a light on it. The light waves will interfere with each other on the other side of the slits, creating a pattern of light and dark bands on the photographic plate (Figure 5.2). There's nothing particularly quantum about this—waves create interference patterns all the time, whether they're overlapping waves from two stones